5.2 Calculate the total charge stored in the channel of an NMOS transistor having $C_{ox} = 25$ fF/μm^2, $L = 65$ nm, and $W = 650$ nm, and operated at $v_{OV} = 0.15$ V and $v_{DS} = 0$ V.

D5.7 An n-channel MOS device in a technology for which oxide thickness is 1.4 nm, minimum channel length is 65 nm, $k_n' = 540\ \mu$A/V^2, and $V_t = 0.35$ V operates in the triode region, with small v_{DS} and with the gate–source voltage in the range 0 V to +1.0 V. If the device has the minimum channel length, what must its width be so that the minimum available resistance is 100 Ω?

5.12 With the knowledge that $\mu_p = 0.25\,\mu_n$, what must be the relative width of n-channel and p-channel devices having equal channel lengths if they are to have equal drain currents when operated in the saturation mode with overdrive voltages of the same magnitude?

5.18 A particular n-channel MOSFET is measured to have a drain current of 360 μA at $v_{GS} = v_{DS} = 1$ V and of 160 μA at $v_{GS} = v_{DS} = 0.8$ V. What are the values of k_n and V_t for this device?

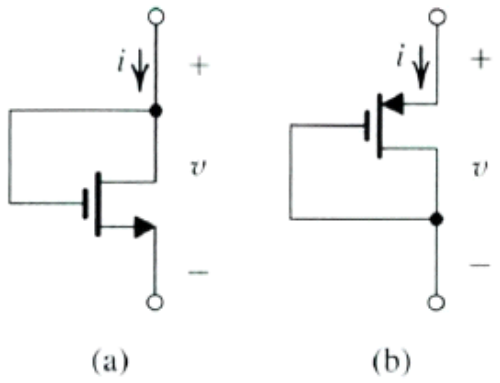
5.23 When the drain and gate of a MOSFET are connected together, a two-terminal device known as a "diode-connected transistor" results. Figure P5.23 shows such devices obtained from MOS transistors of both polarities. Show that

(a) the i–v relationship is given by

$$i = \frac{1}{2}k'\frac{W}{L}(v - |V_t|)^2$$

(b) the incremental resistance r for a device operating at $v = |V_t| + V_{OV}$ is given by

$$r \equiv 1\Big/\left[\frac{\partial i}{\partial v}\right] = 1\Big/\left(k'\frac{W}{L}V_{OV}\right)$$



(a) (b)

Figure P5.23

5.27 The NMOS transistor in Fig. P5.27 has $V_t = 0.5$ V and $k_n = 2$ mA/V^2. Sketch and clearly label i_D versus v_G with v_G varying in the range 0 to +1.8 V. Give equations for the various portions of the resulting graph.

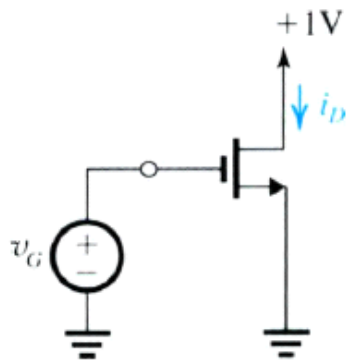


Figure P5.27

D5.31 In a particular IC design in which the standard channel length is 0.18 μm, an NMOS device of standard length with W/L of 10 operating at 100 μA is found to have an output resistance of 20 kΩ, about $\frac{1}{5}$ of that needed. What dimensional change can be made to solve the problem? What is the new device length? The new device width? The new W/L ratio? What is V_A for the standard device in this IC? The new device?

D5.32 For a particular n-channel MOS technology, in which the minimum channel length is 0.2 μm, the associated value of λ is 0.5 V^{-1}. If a particular device for which L is 1.0 μm operates in saturation at $v_{DS} = 1$ V with a drain current of 100 μA, what does the drain current become if v_{DS} is raised to 2 V? What percentage change does this represent? What can be done to reduce the percentage by a factor of 2?

5.44 The NMOS transistor in the circuit of Fig. P5.43 has $V_t = 0.4$ V and $k_n = 4$ mA/V^2. The voltages at the source and the drain are measured and found to be −0.55 V and +0.1 V, respectively. What current I_D is flowing, and what must the values of R_D and R_S be? What is the largest value for R_D for which I_D remains unchanged?

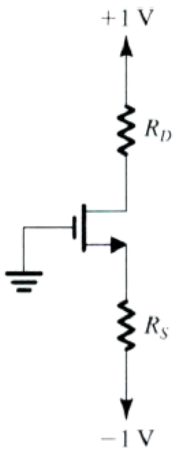


Figure P5.43

D5.47 It is required to operate the transistor in the circuit of Fig. P5.46 at the edge of saturation with $I_D = 50\,\mu A$. If $V_t = 0.4$ V, find the required value of R_D.

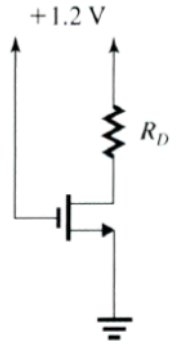


Figure P5.46

5.55 For each of the circuits in Fig. P5.55, find the labeled node voltages. For all transistors, $k_n'(W/L) = 2$ mA/V^2, $V_t = 0.5$ V, and $\lambda = 0$.

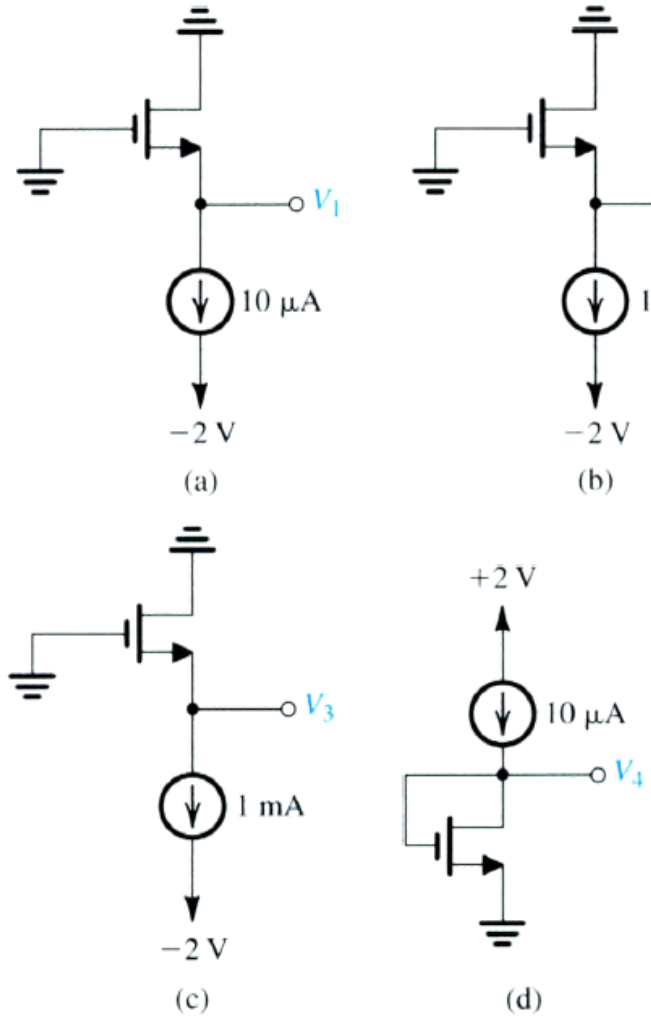


Figure P5.55

***5.56** For each of the circuits shown in Fig. P5.56, find the labeled node voltages. The NMOS transistors have $V_t = 0.4$ V and $k_n'(W/L) = 5$ mA/V^2.

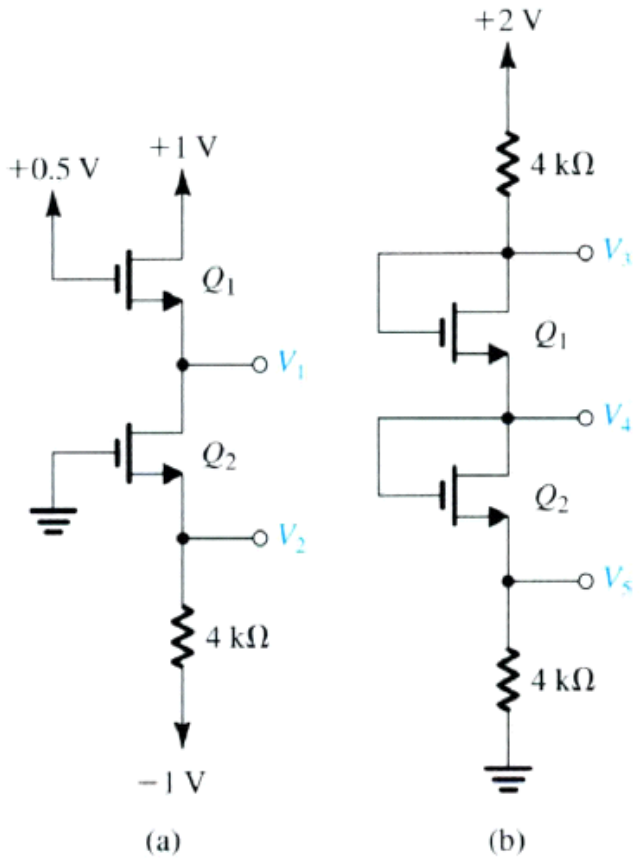


Figure P5.56